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Fm16

Exponentials and Logarithms 4

Review

1 $\sin 3x + 2\cos x$

when x is small, $\sin 3x = 3x$

$2\cos x = 2\left(1 - \frac{x^2}{2}\right)$

$\sin 3x + 2\cos x = -x^2 + 3x + 2$

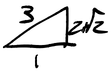
$a = 2,\ b = 3,\ c =$

2 d

3 a is false

4 $\sec x = 3$

$\therefore \cos x = \frac{1}{3}$



$\tan x = \frac{O}{A} = 2\sqrt{2}$

$= \sqrt{8}$ (b)

5 R, P, Q, S

Consolidation

1 $v = 10 + 20e^{-0.2}$

$= 26.4$

$15 = 10 + 20e^{-0.01t}$

$\frac{1}{4} = e^{-0.01t}$

$\ln 4 = 0.01t$

$t = 138\text{s}$

2 $N = 20e^{2}$

$= 148$

$N > 50000$

$20e^{0.2t} > 50000$

$e^{0.2t} > 2500$

$t > 5\ln 2500$

> 39.12

$t = 40$

3 $T = 20 + 80e^{0}$

$= 100°C$

b) $T = 20 + 80e^{-2.5}$

$= 26.6°$

c $80e^{-0.5t} = 10$

$e^{-0.5t} = \frac{1}{8}$

$t = 2(\ln 8)$

$= 4.16$

d) $T' = -160e^{-0.5t}$

$= -160$ when $t = 0$

$= -58.8$ when $t = 2$

e) $20°C$

4 $N = ab^{t}$

$\log N = \log(ab^{t}) = (\log b)t + \log a$

when $t = 0$, $\log N = \log a$

$= 1.9$

$\therefore a = 10^{1.9} \approx 79.4$

$\therefore \log N = t\log b + 1.9$

$2.6 = 4\log b + 1.9$

$0.7 = 4\log b$

$\log b = \frac{7}{40}$

$\therefore b \approx 1.5$

c) $\log N = \frac{7}{40}t + 1.9$

$N = 10^{5.4}$

$= 250{,}000$

maybe not good estimate b/c we are extrapolating

5 $y = kx^{n}$

$\log y = n\log x - \log k$

when $\log x = 0$,

$\log y = -\log k$

$0.08 = \log k$

$k = 10^{0.08}$

$= 1.20$

when $\log y = 0$,

$\log k = n\log x$

$n = \frac{0.08}{0.08}$

$= 1$

b) $y = 10^{n\log x - \log k} = 10^{\log\frac{x}{0.08}}$

$= 780$

unreliable b/c extrapolating outside data range

Exam Qs

1 $T = 25 + ae^{-kt}$

a) $a = 75$

b) 25 is the limit of the temperature of water as $t \to \infty$ (room temp.)

c) $T' = -75ke^{-kt}$

when $t = 0$, $T' = 15$

$75k = 15$
$k = 6$

d) $T = 45$
$25 + 75e^{-6t} = 45$
$-6t = \ln 20 - \ln 75$
$t = 0.22$ mins

e) The value of a would be $T_{water} - 25$; less than 75.

2 a) $y = pq^x$

$\log y = \log p - (\log q)x$

b) when $x = 0$, $\log y = \log p$
$p = 10^{2.6}$

$\log y = \log p - (\log q)x$

$\log 235 = 2.6 - (\log q)x$

$q = 1.694$

c $y < 20$

$\log y < \log 20$

$2.6 - \log 1.69\, x < \log 20$

$x > 5.7$
$x = 6$

unreliable b/c extrapolating outside data range.